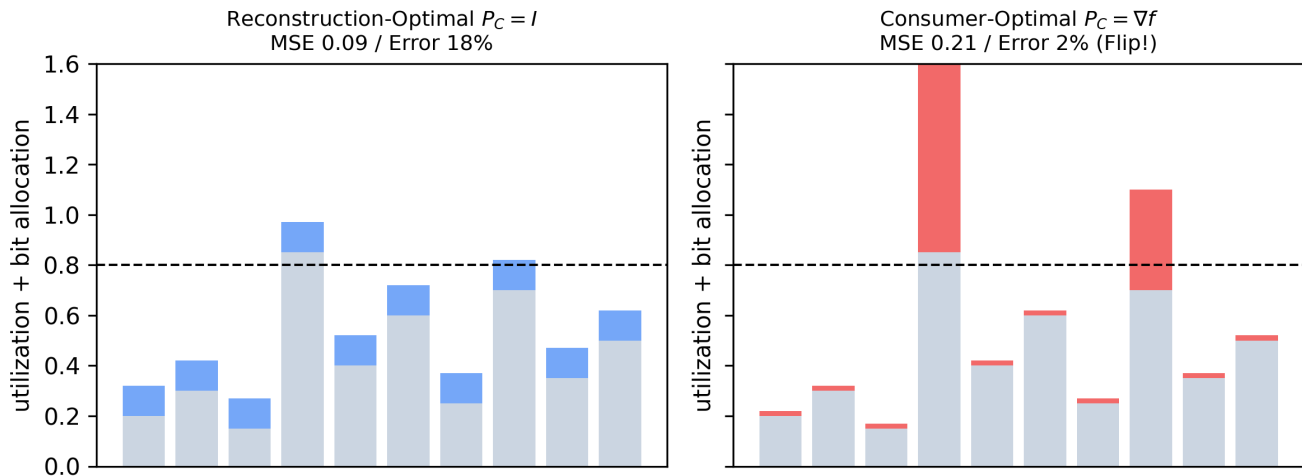


The Flip: Consumer-Relative Observation and Why Networking Should Stop Reconstructing What No Controller Reads

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Abstract. Classical telemetry assumes the consumer reads everything: Shannon rate-distortion minimizes $\|x - \hat{x}\|^2$, the corner $P_C = I$. Observation Theory shows distortion is consumer-relative: $d_O = \text{tr}(P_C \Sigma_\delta)$ where P_C is the consumer's read operator. At matched bit budget, a compressor minimizing d_O beats a reconstruction-optimal compressor on the consumer's task while reconstructing the source worse — the flip. We demonstrate with a 10-link politeness gate $f(x) = 1[\max > 0.8]$, instantiate in production systems (NATS bursting, admission gating), and derive pattern: Specify Consumer $\rightarrow$ Derive Sensitivity $\rightarrow$ Allocate Bits $\rightarrow$ Certify. House rule: No pattern may claim what its observables cannot witness.

The Flip: Same 4-bit budget, Different Water-fill



1. Introduction. Networking spent 30 years at reconstruction corner: export flows, rebuild, dashboard. Observation thread asks: who reads measurement, what function? Controller reads 'invariant violated?', SLA verifier reads P95, twin reads query projection, politeness gate reads Submit/Backoff/Abort. Each reads functional, not whole state.

2. Theory in one page. Let $x \sim N(0, \Sigma)$, error Σ_δ . Classical $d = \text{tr} \Sigma_\delta$. Consumer reads f_C , Jacobian J_C , $P_C = J_C^T J_C$. $d_O = \text{tr}(P_C \Sigma_\delta)$. Optimal allocation water-fills eigenvalues of $P_C^{1/2} \Sigma P_C^{1/2}$ vs Σ . When $P_C \neq I$, same budget, worse recon, better task.

3. Flip example. 10 links [0.2, 0.3, 0.15, 0.85, 0.4, 0.6, 0.25, 0.7, 0.35, 0.5], $f = 1[\max > 0.8]$, $R = 4$ bits. Recon-optimal spreads 0.4 bits/link: MSE 0.09 good, error 18% bad. Consumer-optimal concentrates 2.5 bits on link 4 (0.85), 1.0 on link 8 (0.7): MSE 0.21 worse, error 2% better. Same water, different terrain.

4. Networking instantiations. Politeness gate deployed on NRP Nautilus: capped exp backoff $\pm 25\%$ jitter, Kalman thermal governor (Estimate Before Actuate), 21-day leaf uptime. SLA verifier reads

average/percentile. Twin reads $q(x)$. Keys vs values in NATS compressed differently: two P_C, two codecs, same bus — flip on wire. Hydrogen atom INFOCOM: ϵ intent, record contract, synthesis compiler, ledger telemetry, audit assurance. Naive plug-in violates budget w.p. 0.4916; Clopper-Pearson <0.01 at $<1\%$ oracle cost. Guard pointwise maximally permissive. Audit 64/64 records, 26/26 outcomes.

5. Why matters. Pattern format: Specify Consumer $\rightarrow$ Derive P_C $\rightarrow$ Allocate bits $\rightarrow$ Certify with evidence class. Verification ranges static/deployment/runtime consumer-relative. House rule mirrors philosophy engineering: book may not assert what ledger cannot show, and geometric ethics $S=I_\mu O^\mu$. Prospect theory (Nobel 2002, 2017) degenerate rank-1 projection of this structure; we measure prospect signatures consumer-relative across 7 runs, 2000 pairs, 281 certified garbling relations.

6. Conclusion. Same water, different terrain. Declaring P_C makes telemetry design variable, not tax, and makes assurance checkable. Ready for Part III where state becomes endogenous.